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GATE ECE · SIGNALS AND SYSTEMS · CHAPTER-WISE PYQS

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GATE ECE Signals and Systems · Previous Year Questions with Answer Key

22	15	1995 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	11 questions
Marks 2	11 questions
Answer key	at the end

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Marks 1 11 questions

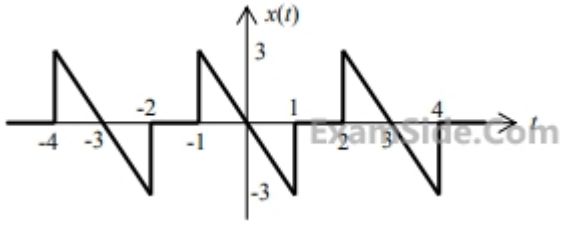
Q1. GATE ECE 2015 Set 3

Consider the function $g(t) = e^{-t} \sin(2\pi t)\, u(t)$ where u(t) is the unit step function. The area under g(t) is_____.

Open solution

Q2. GATE ECE 2015 Set 1

The waveform of a periodic signal x(t) is shown in the figure.



A signal g(t) is defined by $g(t) = x\left(\frac{t-1}{2}\right)$.

The average power of g(t) is_____.

Open solution

Q3. GATE ECE 2014 Set 1

A discrete - time signal x[n] = $\sin(\pi^2 n)$, n being an integer is

(A) periodic with period π
(B) periodic with period π^2
(C) periodic with period is $\pi/2$
(D) not periodic.

Open solution

Q4. GATE ECE 2014 Set 3

The phase response of a passband waveform at receiver is given by
$\varphi(f) = -2\pi\alpha(f - f_c) - 2\pi\beta f_c$
where f_c is the centre frequency, and α and β are positive constants. The actual signal propagation delay from the trasmitter to receiver is

(A) $\frac{\alpha-\beta}{\alpha+\beta}$
(B) $\frac{\alpha\beta}{\alpha+\beta}$
(C) α
(D) β

Open solution

Q5. GATE ECE 2013

For a periodic signal v(t) = 30 sin 100t + 10cos 300t + 6sin $(500t + \pi/4)$, the fundamental frequency in rad/s is

(A) 100 (B) 300 (C) 500 (D) 1500

Open solution

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Q6. GATE ECE 2010

Consider an angle modutated signal $x(t) = 6\ \cos\left[2\pi \times 10^6 t + 2\sin(8000\pi t)\, 4\cos(8000\pi t)\right]$ V. The average power of x(t) is

(A) 10 W (B) 18 W (C) 20 W (D) 28 W

Open solution

Q7. GATE ECE 2006

A solution for the differential equation $\dot{x}$(t) + 2 x (t) = $\delta(t)$ with intial condition $x(0^-) = 0$ is

(A) $e^{-2t}\,u(t)$ (B) $e^{2t}\,u(t)$ (C) $e^{-t}\,u(t)$ (D) $e^{t}\,u(t)$

Open solution

Q8. GATE ECE 2006

The Dirac delta function $\delta(t)$ is defined as

(A) $\delta(t) = \begin{cases} 1, & t = 0 \\ 0, & otherwise \end{cases}$

(B) $\delta(t) = \begin{cases} \infty, & t = 0 \\ 0, & otherwise \end{cases}$

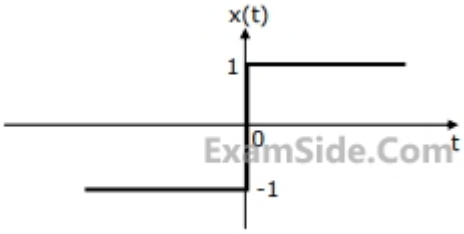
(C) $\delta(t) = \begin{cases} 1, & t = 0 \\ 0, & otherwise\ \ and \int_{-\infty}^{\infty} \delta(t)\,dt = 1 \end{cases}$

(D) $\delta(t) = \begin{cases} \infty, & t = 0 \\ 0, & otherwise\ \ and \int_{-\infty}^{\infty} \delta(t)\,dt = 1 \end{cases}$

Open solution

Q9. GATE ECE 2005

The function x(t) is shown in Fig. Even and odd parts of a unit-step function u(t) are respectively.



(A) $\frac{1}{2}, \frac{1}{2}x(t)$ (B) $-\frac{1}{2}, \frac{1}{2}x(t)$ (C) $\frac{1}{2}, -\frac{1}{2}x(t)$ (D) $-\frac{1}{2}, -\frac{1}{2}x(t)$

Open solution

Q10. GATE ECE 2001

Let $\delta(t)$ denote the delta function. The value of the the integral $\int_{-\infty}^{\infty} \delta(t)\ \cos\left(\frac{3}{2}t\right)dt$ is

(A) 1 (B) - 1 (C) 0 (D) $\pi/2$

Open solution

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Q11. GATE ECE 1998

The ACF of a rectangular pulse of duration T is

(A) a rectangular pulse of duration T
(B) a rectangular pulse of duration 2T
(C) a triangular pulse of duration T
(D) a triangular pulse of duration 2T

Open solution

Marks 2 11 questions

Q12. GATE ECE 2026

Consider a real signal $x(t), -\infty < t < \infty$, such that $x(t) = 0$ for $t < 0, x(t) = 2$ for $0 \leq t < 1$ and $x(t) = 0$ for $t \geq 1$.

Let $E[x(t)] = \int_{-\infty}^{\infty} [x(t)]^2 dt$.

Which of the following options correctly represents the ratio, $E[x(t)]/E[3x(-3t+5)]$?

(A) 3 (B) 1 (C) $1/3$ (D) $1/9$

Open solution

Q13. GATE ECE 2015 Set 3

The complex envelope of the bandpass signal $x(t) = -\sqrt{2}\left(\frac{\sin(\pi t/5)}{\pi t/5}\right)\sin\left(\pi t - \frac{\pi}{5}\right)$, centered about $f = \frac{1}{2}\ Hz$, is

(A) $\left(\frac{\sin(\pi t/5)}{\pi t/5}\right) e^{j\frac{\pi}{4}}$
(B) $\left(\frac{\sin(\pi t/5)}{\pi t/5}\right) e^{-j\frac{\pi}{4}}$
(C) $\sqrt{2}\left(\frac{\sin(\pi t/5)}{\pi t/5}\right) e^{j\frac{\pi}{4}}$
(D) $\sqrt{2}\left(\frac{\sin(\pi t/5)}{\pi t/5}\right) e^{-j\frac{\pi}{4}}$

Open solution

Q14. GATE ECE 2009

An LTI system having transfer function $\frac{s^2+1}{s^2+2s+1}$ and input x(t) = sin (t + 1) is in steady state. The output is sampled at a rate $\omega_s\ rad/s$ to obtain the final output {y(k)}. Which of the following is true?

(A) y(.) is zero for all sampling frequencies ω_s
(B) y(.) is nonzero for all sampling frequencies ω_s
(C) y(.) is nonzero for ω_s > 2, but zero for ω_s < 2
(D) y(.) is zero for ω_s > 2, but nonzero for ω_s < 2

Open solution

Q15. GATE ECE 2007

A Hilbert transformer is a

(A) non-linear system
(B) non-causal system
(C) time-varying system
(D) low-pass system

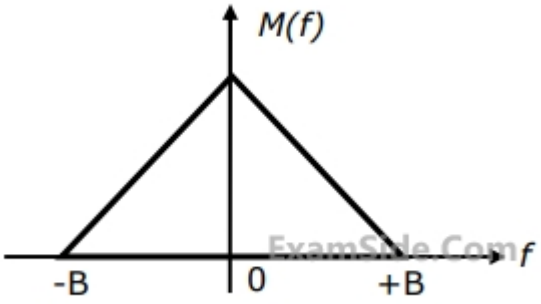
Open solution

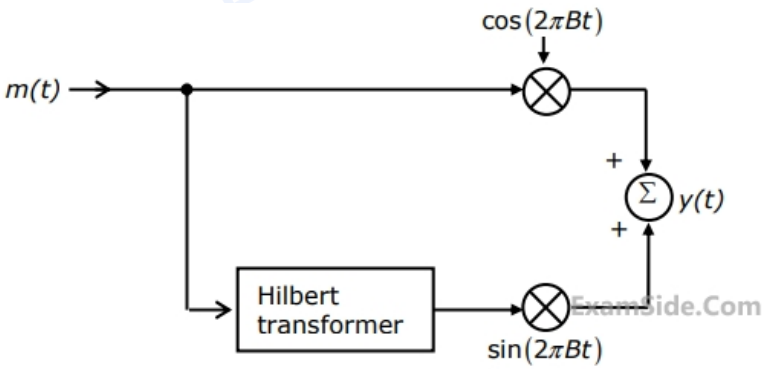
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Q16. GATE ECE 2007

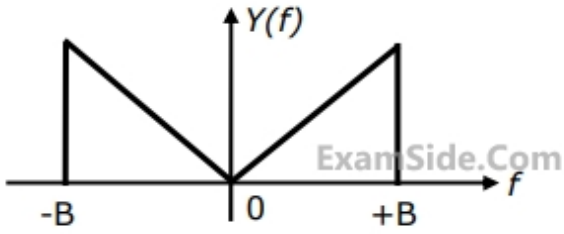
Open solution

In the following scheme, if the spectrum M(f) of m(t) is as shown, then the spectrum Y(f) of y(t) will be:

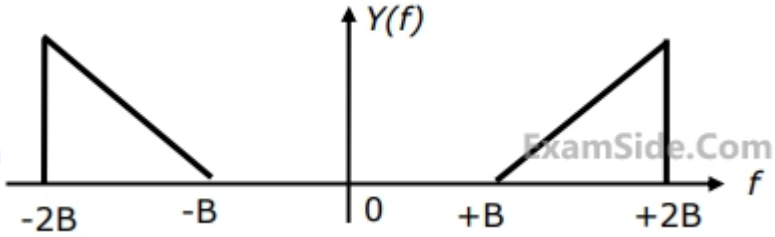




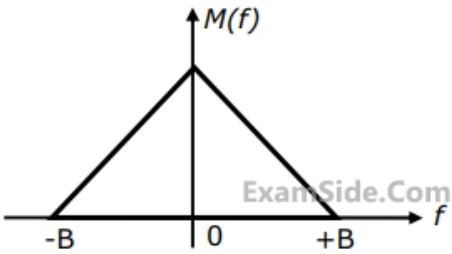
(A)



(B)

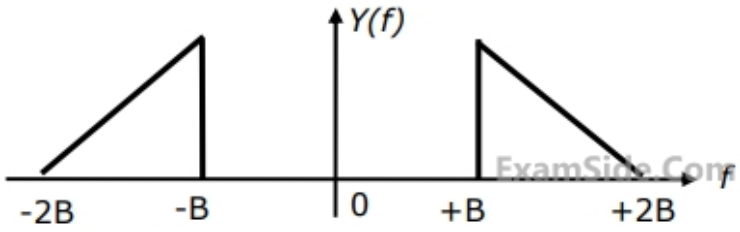


(C)



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(D)



Q17. GATE ECE 2006

As x is increased from $-\infty$ to ∞, the function $f(x) = \frac{e^x}{1+e^x}$

(A) monotonically increases

(B) monotonically decreases

(C) increases to a maximum value and then decreases

(D) decreases to a minimum value and then increases

Open solution

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Q18. GATE ECE 2005

The derivative of the symmetric function drawn in Fig .1 will look like

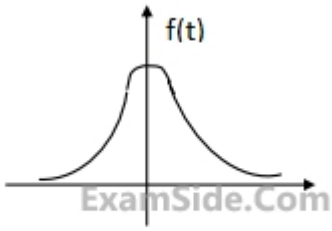
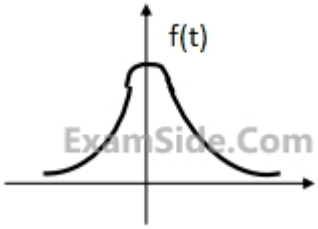
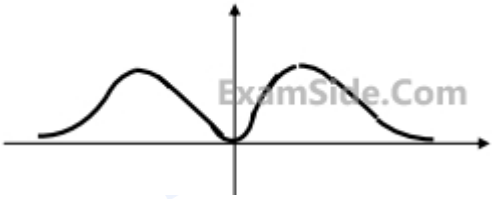


Fig.1

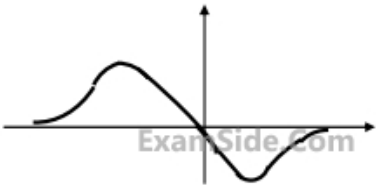
(A)



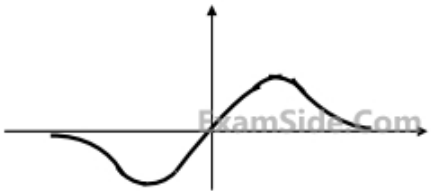
(B)



(C)



(D)



Open solution

Q19. GATE ECE 2005

The value of the integral $I = \frac{1}{\sqrt{2\ \pi}} \int_0^{\infty} \exp\left(-\frac{x^2}{8}\right) dx$ is

(A) 1 (B) π (C) 2 (D) 2π

Open solution

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Q20. GATE ECE 2005

Open solution

Match the following and choose the correct combination.

GROUP 1

E- continuous and aperiodic signal

F- continuous and periodic signal

G- discrete and aperiodic signal

H- discrete and periodic signal

Group 2

1- Fourier representation is continuous and aperiodic

2- Fourier representation is discrete and aperiodic

3- Fourier representation is continuous and periodic

4- Fourier representation is discrete and periodic

(A) E -3, F - 2, G - 4, H -1

(B) E -1, F - 3, G - 2, H - 4

(C) E - 1, F - 2, G - 3, H -4

(D) E -2, F - 1, G - 4, H -3

Q21. GATE ECE 2004

Open solution

Consider the sequence

$x[n] = [-4 - j5,\ 1 + j2,\ \underset{\uparrow}{4}]$

The conjugate anti-symmetric part of the sequence is

(A) $[-4 - j\ 2.5 \quad j2 \quad 4 - j\ 2.5]$

(B) $[-j\ 2.5 \quad 1 \quad -j\ 2.5]$

(C) $[-j\ 5 \quad j\ 2 \quad 0]$

(D) $[-4 \quad 1 \quad 4]$

Q22. GATE ECE 1995

Open solution

Match each of the items, A, B and C, with an appropriate item from 1, 2, 3, 4 and 5

A. Fourier transform of a Gaussian function

B. Convolution of a rectangular pulse with itself

C. Current through an inductor for a step input voltage

1. Gaussian function
2. Rectangular pulse
3. Triangular pulse Ramp function
4. Rump function
5. Zero

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Answer Key

Marks 1

1. **0.15**	2. **2**	3. **D**	4. **C**	5. **A**
6. **B**	7. **A**	8. **D**	9. **A**	10. **A**
11. **D**				

Marks 2

12. **C**	13. **C**	14. **A**	15. **B**	16. **A**
17. **A**	18. **C**	19. **A**	20. **C**	21. **A**
22. **A - 1, B - 3, C - 4**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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